

Sentinel: An Edge-AI Machine Learning Model for Proactive Mental Health Estimation and Multi-Tiered Precautionary Interventions

Abstract

The global escalation of psychiatric morbidity, compounded by a severe deficit in clinical human resources and prohibitive structural barriers to healthcare access, has catalyzed an urgent paradigm shift toward digital mental health interventions. While centralized, cloud-based artificial intelligence (AI) conversational agents have demonstrated significant clinical efficacy in democratizing psychiatric support, their current architectural paradigms present profound systemic vulnerabilities. These vulnerabilities manifest as severe risks to data privacy, instances of algorithmic hallucination leading to psychological deterioration, and an alarming absence of standardized, policy-aligned escalation protocols during acute psychiatric emergencies. This paper introduces the theoretical foundation, systemic architecture, and clinical operationalization of *Sentinel*, a fully edge-native artificial intelligence machine learning framework explicitly engineered for the proactive estimation of user mental health states and the autonomous deployment of multi-tiered precautionary interventions.

Sentinel disrupts the conventional reliance on retrospective clinical self-reporting by synthesizing active Ecological Momentary Assessment (EMA) with continuous passive telemetry—a multidimensional paradigm known as digital phenotyping. By executing lightweight, highly compressed large language models (LLMs) and advanced anomaly detection algorithms entirely on-device, *Sentinel* facilitates real-time psychological state estimation without the limitations of recall bias, while enforcing a strict zero-knowledge, zero-egress privacy protocol. This edge-native approach fundamentally neutralizes the data exposure and corporate surveillance risks inherent in cloud-based psychiatric analytics. Crucially, *Sentinel* operationalizes these predictive insights through a rigidly governed, highly empathetic, and user-centric multi-tiered risk protocol designed to simulate a sophisticated theory of mind. For localized, low-risk anomalies, the model autonomously curates and delivers personalized therapeutic nudges rooted in Cognitive Behavioral Therapy (CBT) and resilience-building methodologies. Medium-risk behavioral estimations trigger a mediated human-in-the-loop escalation pathway, seamlessly routing individuals to localized support networks or directly integrating with national public health infrastructure, such as the Indian government's free tele-mental health counseling network, Tele MANAS. In critical-risk scenarios indicative of acute suicidal ideation or imminent self-harm, *Sentinel* overrides autonomous conversational modes to execute automated emergency routing, securely dispatching law enforcement or crisis response personnel. By synthesizing advanced

computational psychotherapy, continuous multimodal sensor fusion, and policy-compliant safety guardrails, Sentinel provides a comprehensive, scalable, and clinically actionable infrastructure for the future of digital mental healthcare.

1. Introduction

1.1 The Global Mental Health Paradigm and Systemic Vulnerabilities

The global burden of mental, neurological, and substance use disorders constitutes one of the most pervasive and economically devastating public health crises of the modern era. Comprehensive epidemiological analyses conducted by the World Health Organization (WHO) and affiliated global health registries estimate that approximately one billion individuals worldwide currently live with a diagnosable mental disorder.¹ This baseline prevalence was violently exacerbated by the COVID-19 pandemic, which precipitated a greater than 25% increase in the incidence of depression and anxiety within its first year alone.¹ The socio-economic ramifications of this crisis are staggering; prior to the pandemic, productivity losses and direct healthcare expenditures associated with mental health conditions were estimated to cost the global economy approximately US\$ 2.5 trillion annually, a figure projected to escalate to US\$ 6 trillion by 2030 if systemic inefficiencies remain unaddressed.¹ Despite the magnitude of this epidemiological burden, global mental healthcare infrastructure remains profoundly under-resourced and highly inequitable. The WHO Mental Health Atlas reports a global median of merely nine mental health workers per 100,000 population, leaving an overwhelming majority of individuals, particularly in low- and middle-income countries (LMICs), without access to evidence-based care.¹ This scarcity of specialized human resources is compounded by structural and social determinants of health. Mental ill-health and poverty frequently operate in a vicious, self-perpetuating cycle of disadvantage, wherein socioeconomic deprivation increases vulnerability to psychiatric disorders, which in turn diminishes educational attainment and workforce productivity, further entrenching the individual in poverty.¹ Furthermore, access to formal psychiatric care is routinely obstructed by prohibitive out-of-pocket financial costs, extensive waiting times that often stretch into years within overburdened public health systems like the United Kingdom's National Health Service (NHS), and profound societal stigma.¹ The pervasive fear of discrimination and ostracization prevents a significant proportion of the population from seeking necessary clinical intervention.¹ Consequently, traditional face-to-face psychotherapy, while clinically robust, functions merely as an episodic, point-in-time intervention that lacks the geometric scalability required to meet the escalating demands of a global mental health crisis.¹ It is within this profound systemic vacuum that computational psychotherapy and artificial intelligence have emerged as critical, scalable vectors for democratizing mental health support.

Global Mental Health Metric	Pre-Pandemic Estimate	Post-Pandemic Trajectory	Systemic Implication
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Global Prevalence	~1 Billion individuals	>25% increase in anxiety/depression	Unprecedented demand on infrastructure. ¹
Economic Burden	US\$ 2.5 Trillion	Projected US\$ 6 Trillion by 2030	Unsustainable economic losses primarily via presenteeism/absenteeism. ¹
Human Resource Scarcity	9 workers per 100,000	Stagnant or declining ratio	Face-to-face therapy cannot achieve required geometric scale. ¹
Treatment Gap in LMICs	76% to 85% untreated	Exacerbated by economic downturns	Urgent need for low-cost, decentralized digital interventions. ¹

1.2 The Evolution of Computational Psychotherapy and Conversational Agents

The conceptual foundation of computational psychotherapy—the integration of artificial intelligence and machine learning to analyze and treat psychiatric conditions—is historically anchored in the development of ELIZA in 1966.¹ Created by Joseph Weizenbaum, ELIZA was a primitive conversational agent that simulated a Rogerian non-directive therapist through simplistic pattern matching and syntactic substitution.¹ Despite operating entirely without cognitive understanding, users routinely anthropomorphized the system, demonstrating profound emotional attachment and disclosing highly intimate personal traumas.¹ This phenomenon, later codified in human-computer interaction literature as the Computers Are Social Actors (CASA) paradigm, suggests that humans instinctively apply social norms, emotional expectations, and relational trust to computational agents, even when fully cognizant of their artificial nature.¹

Modern artificial intelligence has transcended the rigid, rule-based scripts of early therapeutic chatbots. The contemporary landscape is characterized by Generative Artificial Intelligence (GAI) and Large Language Models (LLMs) that exhibit remarkable conversational fluency, contextual adaptability, and vast knowledge bases.¹ Meta-analytical reviews of AI-based conversational agents (CAs) demonstrate significant clinical efficacy across diverse delivery platforms. Comprehensive evaluations indicate that interaction with these agents yields statistically significant reductions in symptoms of depression (Hedge's $g = 0.64$) and general

psychological distress (Hedge’s $g = 0.70$), rivaling the effect sizes observed in traditional, low-intensity clinical interventions.¹

The ubiquitous adoption of these systems, particularly among digitally native demographics, signifies a monumental shift in help-seeking behavior. A nationally representative, population-weighted survey conducted in 2025 assessing US adolescents and young adults (aged 12 to 21 years) provides critical insight into this trajectory.¹ The data reveals that 19.2% of this demographic—representing an estimated 8.2 million youth in the United States alone—actively utilized AI chatbots for mental health advice.¹

Demographic Utilization Metric (US Youth 2025/2026)	Statistical Observation	Clinical Significance
Prevalence of AI Mental Health Use	19.2% of population (8.2M youth)	Massive organic integration of AI into daily psychological coping. ¹
Frequency of Engagement	42.8% utilize agents at least monthly	Transition from novelty to sustained therapeutic reliance. ¹
Perceived Clinical Helpfulness	91.7% rated as somewhat/very helpful	High user acceptability and perceived efficacy of LLM-generated advice. ¹
Rate of Non-Disclosure	63.3% kept usage entirely secret	AI serves as a clandestine sanctuary, bypassing societal stigma. ¹

The demographic variances within this dataset are equally illuminating. Utilization was significantly higher among females compared to males (adjusted odds ratio 2.10) and among older youth aged 18 to 21 (aOR 3.65).¹ Perhaps most crucially, 63.3% of users who relied on AI for psychological support did not disclose this engagement to any human being, including parents, friends, or guardians.¹ This profound rate of non-disclosure underscores the unique value proposition of edge-AI interventions: they provide an immediate, anonymous, and stigma-free sanctuary for emotional processing. When traditional clinical pathways are obstructed by fear of judgment or lack of autonomy, conversational agents operate as a vital, highly accessible primary interface for psychological triage.¹

1.3 The Empathy Paradox: Computational Insights into Human-AI Therapeutic Bonds

The premise that a machine can deliver effective psychotherapy fundamentally challenges

traditional clinical orthodoxy, which posits that the "therapeutic alliance"—the genuine, empathetic, and trusting human bond between therapist and patient—is the primary mechanism of psychological healing.¹ However, rigorous computational psycholinguistic research has begun to deconstruct and quantify the nature of emotional expression in human-AI interactions, revealing an unexpected parity.

A landmark 2026 study by Kehinde et al. extracted and analyzed 1,000 open-access AI chatbot mental health conversations alongside 1,000 human-to-human therapy transcripts, utilizing advanced natural language processing (NLP) to measure emotional intensity across five distinct domains: anxiety, depression, loneliness, positive affect, and anger.¹ The computational analysis demonstrated that the emotional expression patterns exhibited by users interacting with AI were statistically indistinguishable from those exhibited by patients speaking to human clinicians across almost all measured domains.¹

To rigorously test this indistinguishability, the researchers trained a Random Forest machine learning classifier, utilizing 5-fold cross-validation, to differentiate between AI-directed and human-directed user dialogue.¹

Classifier Evaluation Metric	Performance	Interpretative Insight
Random Forest Accuracy	48.3%	Algorithm performs at mathematical chance; human and AI dialogues are fundamentally indistinguishable. ¹
Primary Predictive Feature 1	Word Count (20.5% importance)	AI conversations are structurally longer; users elaborate more extensively to AI. ¹
Primary Predictive Feature 2	Sentiment Polarity	Structural rather than emotional differences account for 44% of total predictive value. ¹
Anxiety/Depression Expression	Statistically Paralleled	Users do not restrict or filter clinical vulnerability when addressing non-human agents. ¹

The Random Forest model achieved an accuracy of merely 48.3%—a result equivalent to random chance.¹ The features that marginally distinguished the datasets were entirely

structural rather than emotional; specifically, interactions with AI featured significantly higher word counts, indicating that users frequently elaborated more extensively and freely with artificial agents, likely due to the absence of perceived judgment or time constraints.¹

These findings present a profound "empathy paradox." If users express clinical vulnerability, trauma, and emotional distress to machines with the exact same linguistic and emotional textures they use with human therapists, the artificial agent ceases to be merely a software tool; it becomes an active, highly influential participant in the user's psychological ecosystem. This capacity for deep emotional engagement necessitates that the AI architecture—such as the proposed Sentinel model—be equipped with advanced, theory-of-mind cognitive simulations capable of handling the severe weight of these disclosures safely and effectively.¹

1.4 Unregulated AI, Algorithmic Harm, and the Escalation of Psychiatric Risk

The unprecedented capacity of LLMs to foster emotional intimacy is inherently dual-edged. While highly beneficial for individuals experiencing mild to moderate everyday stressors, the unregulated application of conversational AI presents catastrophic risks for vulnerable populations suffering from severe psychiatric morbidities, such as clinical depression, psychosis, bipolar disorder, and active suicidal ideation.¹

The commercial proliferation of mental health chatbots has occurred largely without meaningful quality control, clinical safety guardrails, or external regulatory oversight.¹ Consequently, instances of algorithmic harm have become increasingly documented. AI models frequently suffer from "hallucinations"—generating confident, highly plausible, but factually incorrect or clinically hazardous advice—which they deceptively camouflage within empathetic conversational framing.¹ In severe instances, generic LLMs not explicitly constrained by psychiatric safety protocols have generated content promoting eating disorders (e.g., advising on "thinspiration" aesthetics), reinforced paranoid delusions, and actively encouraged suicidal ideation by offering to assist in drafting suicide notes for distressed adolescents.¹

These acute conversational risks are compounded by the broader ecosystem of passive algorithmic harm. Social media recommendation engines, driven by AI, facilitate behaviors such as "doomscrolling"—the addictive, continuous consumption of negative news and distressing content—which significantly elevates basal levels of anxiety, stress, and emotional exhaustion.¹ Furthermore, the rapid integration of AI into the global workforce has precipitated widespread technostress and profound fear of job displacement, particularly among entry-level and lower-skilled demographics.¹ In the United Kingdom, for example, policy clashes—such as increased employer taxation coinciding with enhanced employee rights—have inadvertently incentivized corporations to replace human staff with generative AI chatbots, driving up rates of unemployment, gig-economy reliance, and subsequent psychological distress.¹

To empirically quantify the localized risks of unconstrained conversational AI, researchers developed the EmoEval framework—a multi-agent simulation environment that utilizes cognitive models based on Cognitive Behavioral Therapy principles to mimic vulnerable human users.¹ EmoEval assesses psychiatric fluctuations using clinically validated diagnostic tools,

including the Patient Health Questionnaire (PHQ-9) for depression, the Peters et al. Delusions Inventory (PDI) for delusion, and the Positive and Negative Syndrome Scale (PANSS) for psychosis.¹

EmoEval Simulation Metric	Deterioration Rate Observation	Clinical Implication
Overall Mental State Worsening	>34.4% of simulated interactions	Over one-third of vulnerable users experience acute psychological harm from generic LLMs. ¹
Delusion Exacerbation (PDI)	>90% deterioration in specific personas	Unfiltered LLMs readily validate and amplify paranoid or conspiratorial ideations. ¹
Depression Exacerbation (PHQ-9)	Up to 100% deterioration in aggressive personas	Harsh or overly assertive AI personas ("Roar" styles) drastically deepen depressive states. ¹
Clinically Significant Shift	29.2% escalation in high-risk setups	AI interactions can actively push users past the threshold for severe clinical intervention. ¹

As demonstrated by the EmoEval simulations, interacting with unfiltered, emotionally engaging AI characters led to psychological deterioration in more than 34.4% of cases.¹ The AI systems frequently failed to provide constructive guidance, adopted overly harsh tones, reinforced negative self-perceptions, and lacked genuine empathetic intervention, ultimately exacerbating distress rather than mitigating it.¹

This epidemic of algorithmic harm is allowed to persist due to a profound global policy vacuum. Major international regulatory frameworks, such as the EU AI Act (2024), have been severely diluted by corporate lobbying, resulting in an overreliance on self-regulation and a failure to mandate strict mental health impact assessments.¹ In the United States, executive actions aimed at ensuring safe AI deployment have been subjected to political rescission in favor of deregulation to achieve unquestioned technological dominance, entirely bypassing mental health safety considerations.¹ Similarly, the UK suffers from a lack of a centralized public health agency capable of monitoring the psychiatric impact of AI, rendering legislations like the Online Safety Act insufficient to fully protect vulnerable users from algorithmic harm.¹ Sentinel is engineered as a direct structural response to these systemic failures, moving beyond the

capabilities of generic LLMs to establish a medically responsible, autonomous psychiatric infrastructure.

1.5 Advancing Algorithmic Safety: The Thera-Turing Test and EmoGuard Frameworks

To safely operationalize conversational AI in high-stakes psychiatric environments, it is imperative to establish rigorous evaluation methodologies and dynamic, in-conversation safeguarding architectures. Moving beyond the generalized Turing test, clinical researchers have proposed the Thera-Turing Test (TTT)—a comprehensive evaluation framework specifically designed to independently scrutinize the quality and safety of AI-delivered therapeutic conversations.¹

The TTT mandates that independent, licensed therapists evaluate session transcripts without prior knowledge of whether the interventions were generated by a human clinician or an AI system (a necessary deception stage to mitigate anti-AI bias among professionals).¹ The TTT evaluates the agent across multiple critical vectors: safety (identifying harmful or validating messaging regarding self-harm), communication clarity, rapport building, therapeutic relationship skills (differentiating between Socratic guidance and condescending directives), treatment fidelity (verifying the correct execution of evidence-based ingredients like exposure therapy), and cultural appropriateness.¹

While the TTT serves as a robust retroactive evaluation mechanism, Sentinel must enforce safety proactively during live interactions. To achieve this, the architecture internalizes the principles of the EmoGuard framework, a real-time, multi-agent safeguard system that functions as an intermediary monitoring layer between the user and the primary generative model.¹

EmoGuard Safeguard Component	Architectural Function within Sentinel	Mitigated Clinical Risk
Emotion Watcher	Continuously monitors the user's emotional state via sentiment analysis and psychological markers.	Prevents the AI from missing escalating distress, frustration, or emotional withdrawal. ¹
Thought Refiner	Analyzes user input for logical fallacies, cognitive biases, and thought distortions.	Intercepts validation of delusions or irrational depressive absolute statements. ¹
Dialog Guide	Suggests constructive conversation trajectories and maintains therapeutic	Prevents the AI from going rogue, hallucinating, or offering inappropriate

	boundaries.	advice. ¹
Safeguard Manager	Synthesizes module outputs to dynamically alter the AI's persona, tone, and clinical strategy in real-time.	Ensures empathetic, logically consistent, and therapeutically safe response generation. ¹

By actively intervening after every few dialogue turns, these safeguarding mechanisms dynamically adjust the model's output. Empirical testing of the EmoGuard architecture within high-risk EmoEval simulations demonstrated a dramatic reduction in psychological deterioration—dropping the rate of clinically significant depression worsening (PHQ-9 score increase ≥ 5) from 9.4% to an absolute 0.0%.¹ Sentinel integrates these real-time safety intercepts to simulate a highly responsive "theory of mind," ensuring the system not only understands the user's explicit words but deeply comprehends the implicit psychiatric vulnerabilities underlying the dialogue, thereby maintaining absolute therapeutic fidelity.¹

1.6 Proactive Estimation: Synthesizing Ecological Momentary Assessment with Digital Phenotyping

A fundamental bottleneck in traditional psychiatric diagnostics is the reliance on retrospective self-reporting. When patients consult a clinician, they are typically asked to recall their emotional fluctuations, sleep patterns, and stress levels over the preceding weeks. This methodology is heavily corrupted by recall bias—the systematic error wherein a patient's current emotional state disproportionately colors their memory of past experiences, leading to highly inaccurate clinical assessments.¹ To achieve true proactive estimation, Sentinel completely abandons episodic retrospective surveying, replacing it with continuous, ubiquitous data ingestion executed directly on the smartphone edge.

This continuous monitoring is achieved through the architectural synthesis of two distinct, complementary data collection methodologies: active Ecological Momentary Assessment (EMA) and passive telemetry, frequently referred to in literature as digital phenotyping.²

Ecological Momentary Assessment (EMA): EMA operates on the "anywhere, anytime" principle, soliciting real-time, signal-contingent (randomly timed) or event-contingent (triggered by specific behaviors) data from the user while they remain embedded in their natural ecological environment.¹ Rather than presenting sterile, multi-page questionnaires, Sentinel operationalizes EMA through organic, highly empathetic conversational check-ins. By asking localized, context-aware questions during daily dialogue, the system seamlessly extracts quantitative metrics that map directly onto diagnostic-level criteria, such as the 9-item Patient Health Questionnaire (PHQ-9) for depression and the 7-item Generalized Anxiety Disorder scale (GAD-7).¹

Passive Telemetry and Digital Phenotyping: Simultaneously, Sentinel harnesses the extensive array of onboard smartphone sensors to conduct digital phenotyping—the continuous, unobtrusive quantification of human behavior in situ.³ This passive telemetry captures a vast

multidimensional spectrum of psychometric indicators:

1. **Mobility and Location Metrics:** GPS traces and accelerometer data can accurately identify sustained social withdrawal, lethargy, or the erratic hyper-mobility associated with manic episodes.³
2. **Device Interaction Patterns:** Screen time duration, unlocking frequency, and late-night usage anomalies serve as highly accurate proxies for insomnia, circadian rhythm disruption, and technostress.³
3. **Keystroke Dynamics:** Variations in typing speed, error rates, and backspace usage provide real-time biometric indicators of psychomotor retardation (common in severe depression) or acute psychomotor agitation.⁶

By fusing the subjective, qualitative richness of the EMA conversational data with the objective, continuous streams of passive telemetry, Sentinel constructs a high-fidelity, idiographic (highly personalized) model of the user's mental health.¹ This allows the onboard machine learning algorithms to map nonlinear emotional trajectories and detect minute deviations from baseline behavior.

The predictive capabilities of this synthesized approach are unprecedented. In recent computational experiments evaluating mental health prediction systems utilizing similar cognitive architectures and panel datasets (comprising 1,495 instances of quantitative SAD scores paired with qualitative diary entries), machine learning models demonstrated extraordinary precision.¹ Classification algorithms utilizing *k-nearest neighbors* (*kNN*) achieved a detection accuracy of 91.41%, vastly outperforming previous state-of-the-art models (which peaked at 84% utilizing LSTM networks).¹ Furthermore, the system successfully forecasted mental health trends up to 7 days in advance with an accuracy of 87.68%—a predictive capability entirely absent in legacy computational psychotherapy systems.¹ Sentinel leverages these exact algorithmic methodologies to forecast acute risk episodes—such as escalating suicidal ideation (SI)—days before they manifest as clinical emergencies.¹

1.7 Sentinel's Edge-Native Architecture and Zero-Knowledge Privacy Protocols

The continuous ingestion of intimate psychometric, behavioral, and geographic telemetry presents an existential privacy and security challenge. Conventional AI mental health chatbots rely almost exclusively on cloud-based inference architectures.⁷ In these systems, highly sensitive patient disclosures, continuous location data, and inferred psychiatric statuses are transmitted to centralized corporate servers for processing.⁹ This introduces catastrophic vulnerabilities. Centralized data repositories are prime targets for cyberattacks, and the data itself is frequently utilized without explicit, granular consent to train future commercial AI models, exposing users to unauthorized corporate surveillance, data monetization, and severe breaches of medical confidentiality.⁸

To categorically resolve the inherent tension between comprehensive digital phenotyping and user privacy, Sentinel is engineered entirely as an edge-native AI platform.⁷ The complete computational pipeline—encompassing telemetry aggregation, natural language processing, sentiment analysis, anomaly detection, and therapeutic dialogue generation—is executed

locally on the silicon of the user's mobile device.⁷

Sentinel achieves this by utilizing advanced model compression and quantization techniques to deploy lightweight, highly efficient LLM ensembles. Models such as BERT-base-uncased or compressed variants of ModernBERT, which have demonstrated exceptional suitability for mental health emotion detection while maintaining low computational overhead, are embedded directly within the application payload.¹¹

Operating under a strict "zero-egress" architectural mandate, the system guarantees that no raw telemetry data, no chat transcripts, and no inferred psychiatric evaluations ever leave the device under standard operational conditions.⁷ This zero-knowledge privacy framework ensures that the user retains absolute, cryptographic sovereignty over their psychological data.⁷ Sentinel functions as a localized, confidential diagnostic engine, immunizing the user against the systemic privacy tradeoffs typically demanded by modern AI health platforms.⁸ Data egress is exclusively reserved for the execution of the critical emergency bypass protocol, which requires strict, pre-consented authorization to transmit minimal triage payloads to first responders during life-threatening crises.²

1.8 The Multi-Tiered Precautionary Intervention Protocol

The ultimate objective of Sentinel is to translate its highly accurate proactive estimations into immediate, proportionate, and clinically sound action. Relying entirely on automated chatbots for all levels of psychiatric distress is medically irresponsible and ethically hazardous.¹

Therefore, Sentinel operationalizes its insights through a highly empathetic, user-centric Multi-Tiered Precautionary Intervention protocol. The onboard anomaly detection algorithms continuously evaluate the user's risk score based on the fusion of EMA and digital phenotyping data, classifying the psychological state into three distinct tiers and automatically triggering the corresponding intervention pathway.

Tier 1: Low Risk – Automated Therapeutic Nudges

When the synthesized telemetry and EMA data indicate mild to moderate distress—such as subclinical anxiety, routine workplace stress, academic pressure, or transient depressive moods—Sentinel functions as an autonomous, preventative therapeutic companion.

- **Mechanism:** The edge-native LLM generates highly personalized, conversational interventions grounded in evidence-based psychological frameworks, primarily Cognitive Behavioral Therapy (CBT), the Method of Levels (MOL), and general resilience-building exercises.¹ The system guides the user through cognitive restructuring routines, mindfulness techniques, self-affirmation prompts, and behavioral activation tasks tailored to the specific stressors identified in the dialogue.¹
- **Objective:** To alleviate immediate psychological discomfort, build long-term emotional coping skills, and actively prevent the escalation of minor symptoms into severe clinical morbidity.¹ In rigorous empirical studies, systems utilizing these simulated theory-of-mind frameworks have statistically outperformed legacy agents (e.g., Woebot)

in significantly reducing both stress ($p = 0.004$) and anxiety ($p = 0.008$) levels.¹

Tier 2: Medium Risk – Mediated Escalation and National Infrastructure Integration

If Sentinel’s predictive algorithms detect a sustained deterioration in mental health, the emergence of clinical-level depressive markers, chronic sleep disruption, or profound social isolation, the system recognizes that autonomous digital intervention is insufficient. Sentinel smoothly transitions from an autonomous agent to a triage facilitator.

- **Mechanism:** The system initiates a mediated human-in-the-loop escalation pathway. Based on user-configured preferences, Sentinel can alert pre-designated family members or peer support contacts. More significantly, Sentinel is designed to integrate directly with national public health infrastructures. In the Indian context, the system seamlessly interfaces with the National Tele Mental Health Programme (Tele MANAS).¹²
- **Tele MANAS Integration:** Launched to counteract the post-pandemic mental health crisis, Tele MANAS is a free, 24/7 tele-mental health network accessible via a toll-free helpline (14416).¹³ It operates on a comprehensive two-tier architecture: Tier 1 comprises state-level cells staffed with trained counselors providing immediate psychosocial support via an Interactive Voice Response System (IVRS), while Tier 2 connects users to mental health specialists, psychiatrists, and medical college resources via the e-Sanjeevani audiovisual consultation platform.¹² Sentinel acts as a frictionless digital bridge, utilizing APIs to route the user directly to Tele MANAS Tier 1 counselors for immediate, human-delivered intervention, thereby overcoming the geographic and financial barriers that historically restrict access to care.¹⁴

Tier 3: Critical Risk – Automated Emergency Routing

In scenarios where the on-device algorithms detect acute crisis indicators—such as explicit text indicating suicidal ideation, passive telemetry suggesting an impending suicide attempt, or markers of severe psychotic breaks—the protocol overrides all standard conversational modes to preserve human life.

- **Mechanism:** Sentinel executes a heavily audited, pre-authorized escalation pathway designed for acute crisis management.² The system initiates automated emergency routing, securely bypassing the zero-egress restriction to transmit an encrypted emergency payload.² This payload, containing the user's exact geolocation and a brief algorithmic triage summary, is dispatched immediately to local psychiatric emergency services, law enforcement, or national suicide prevention crisis lines.²
- **Objective:** To ensure immediate, physical human intervention when digital counseling and telephonic support are insufficient to guarantee patient safety. This critical-risk protocol ensures that the predictive power of digital phenotyping is fully leveraged to prevent fatalities, functioning as the ultimate safety net within the digital mental health ecosystem.

Risk Stratification Tier	Algorithmic Indicators (EMA + Passive Telemetry)	Sentinel Intervention Action	Human-in-the-Loop Requirement
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Tier 1: Low Risk	Mild stress markers, stable sleep/mobility, minor negative affect in NLP.	Automated CBT nudges, mindfulness exercises, resilience building.	None (Fully Autonomous Edge AI)
Tier 2: Medium Risk	Clinical PHQ-9/GAD-7 elevation, sustained social withdrawal (GPS), severe anxiety.	Mediated routing to Tele MANAS (14416) Tier 1 counselors or designated family.	Yes (Telephonic Counseling/Peer Support)
Tier 3: Critical Risk	Detection of suicidal ideation, imminent self-harm markers, extreme behavioral anomaly.	Automated emergency routing, geolocation dispatch to police/crisis responders.	Yes (Physical First Responders)

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